

Using Large-Language-Models in Application Programs

LLM as Image Analysis Agent

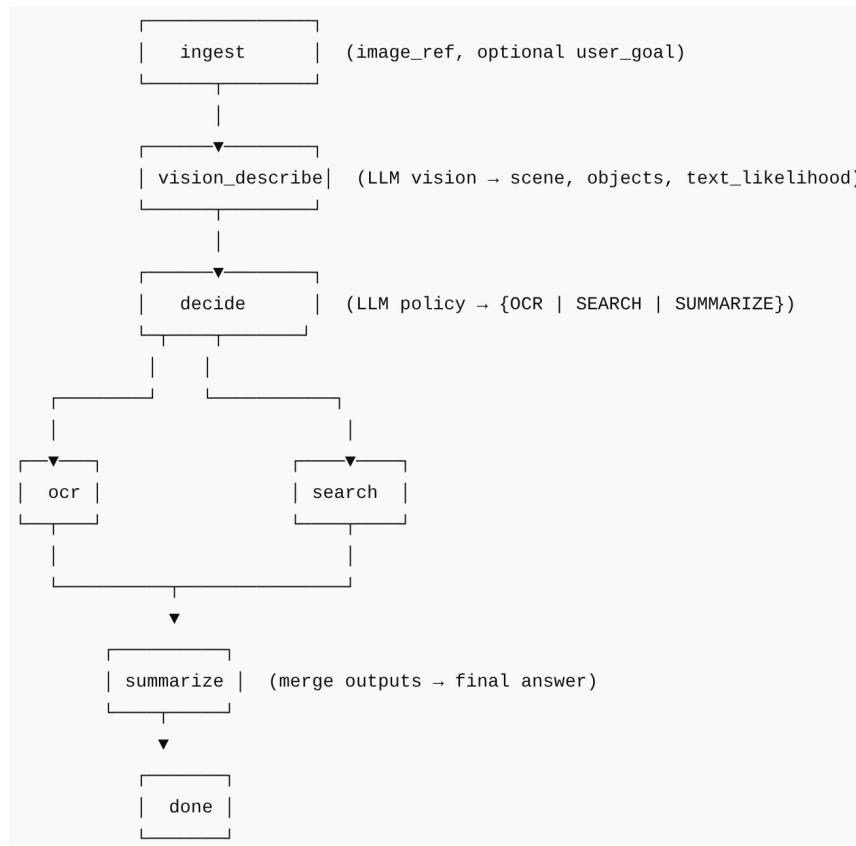
- Current LLMs can describe images.
- We want to build an own program that gets access to our own images and describes it.
- We want to differentiate between a few use cases:
 - Family image
 - Product image
 - Optical character recognition application

Example: Visual Task Helper

- Flow:
 - Take an image (local file or webcam)
 - Vision model describes it.
 - Agent decides whether to:
 - Summarize the scene
 - OCR and extract text
 - Call a secondary tool (e.g., a search API) with detected objects
- Integrates
 - LangGraph branching logic
 - Vision models
 - Tool call

LangGraph and LangChain

- LangGraph is an open-source orchestration framework for building LLM/agent workflows as a graph



- LangChain is an open-source framework for building LLM-powered apps and agents.

Generate an LLM key

- Register and add a one-time \$5 (prepaid) balance
 - 1) **Create / sign in** to the OpenAI developer platform.
 - 2) Open **Settings** → **Billing (Billing overview)** and click “**Add payment details**”.
 - 3) Choose **prepaid credits** and select **\$5** as the initial purchase (minimum), then confirm.
- Create the API key and put it in .env
 - 4) Go to your org/project **API Keys** page and click **Create new secret key** (optionally set it to Restricted permissions).
 - 5) Add to your .env: `OPENAI_API_KEY="sk-...your key..."`
- Keep it out of git / public repos.

Run Visual Task Agent

- Download and unpack vision-agent.zip file.
- Generate conda environment:

```
conda env create -f environment.yml  
conda activate vision-agent
```

- Store dotenv as .env and enter API key.
- Run:

```
python main.py
```

```
(vision-agent) hellwich@Artemis:~/olaf/LLMWS/vision-agent$ python main.py  
vision-agent CLI  
- Paste an image path/URL/data:... to add it as a resource  
- Type a goal sentence to run on the current resources  
- Commands: list | reset | pop | help | quit  
  
vision-agent> ../Urgrossvater2_001.jpg  
added image (1): ../Urgrossvater2_001.jpg  
  
vision-agent> https://www.fradashop.de/thumbnail/cc/4a/53/1704373537/tischler_1280x1280.png?ts=1704373539  
added image (2): https://www.fradashop.de/thumbnail/cc/4a/53/1704373537/tischler_1280x1280.png?ts=1704373539  
  
vision-agent> What do the two images have in common?  
  
[protocol] decision=SUMMARIZE  
  
=== FINAL ANSWER ===  
  
Both images share a common theme of traditional craftsmanship and community.  
  
1. **Historical Context**: Image 1 features a group of people in a rural setting, likely engaged in community or family activities, while Image 2 showcases a man focused on woodworking, highlighting the importance of skilled trades in historical contexts.
```